
Do Agents Learn Pragmatics, or Just Their Partner?

Cross-Play Evaluation of Communicative Self-Play in Situated Reference Games

Anonymous authors

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Abstract

1 Interactive language agents are often evaluated with the same simulated
2 partners that provide feedback or selection signals. In communicative set-
3 tings, this can conflate robust pragmatic competence with partner-specific
4 conventions. We study this issue in PRAG-CrossPlay, a controlled situated
5 reference game where a speaker must produce a natural-language mes-
6 sage that lets a listener identify a hidden target object. Because listener
7 choices are scored against ground-truth object IDs, the task gives an ob-
8 jective communicative-success signal without an LLM judge. In a hybrid
9 API experiment, an API speaker generates candidate messages, a fixed
10 simulated partner or a small listener population selects among them, and
11 held-out API listeners evaluate the selected messages. Mirror self-play
12 achieves perfect same-partner success but drops under cross-play, espe-
13 cially on perspective-sensitive references. In a 50-scene perspective stress
14 replication, mirror self-play obtains 1.000 same-play success but only 0.813
15 cross-play success, while population-play reaches 1.000 cross-play success
16 on the same candidate sets; with a different held-out model family, mir-
17 ror drops further to 0.713 while population-play remains at 1.000. A no-
18 coordinate ablation shows that this result depends on the availability of
19 explicit listener-invariant fallbacks, and that selector quality matters when
20 those fallbacks are removed. These results suggest that cross-play should
21 be a standard diagnostic for language agents trained or selected through
22 situated interaction.

23 1 Introduction

24 Pragmatic communication is partner-sensitive: speakers choose utterances with respect
25 to what a listener can observe, infer, and disambiguate. Interactive language-agent work
26 increasingly uses simulated interaction to improve or evaluate social behavior, including
27 broad social environments such as SOTOPIA [Zhou et al. \(2023\)](#) and interactive-learning
28 variants such as SOTOPIA- π [Wang et al. \(2024\)](#). This creates a useful but risky feedback
29 loop. If the partner that supplies feedback has stable quirks, high interaction success may
30 reflect adaptation to that partner rather than robust pragmatic competence.

31 A related warning appears in zero-shot coordination: self-play can induce specialized con-
32 ventions that do not transfer to independently trained or novel partners, motivating cross-
33 play and inter-algorithm cross-play evaluations [Hu et al. \(2020\)](#); [Lucas & Allen \(2022\)](#).
34 PRAG-CrossPlay brings this diagnostic view to natural-language pragmatic communica-
35 tion. Instead of measuring team reward in a game such as Hanabi, we measure whether a
36 message lets held-out listeners recover a ground-truth referent.

37 We test this issue in situated reference games, a classic setting for studying pragmatic in-
38 ference and informative utterance choice [Frank & Goodman \(2012\)](#); [Hu et al. \(2021\)](#). A
39 speaker sees a target object in a small grid scene and sends a natural-language message
40 to a listener, who must choose the intended object from the listener-visible scene. The re-
41 sulting success signal is objective: the listener either chooses the target object ID or not.
42 This lets us separate same-partner success from cross-partner success without relying on
43 an LLM judge.

Our core question is simple: when a speaker selects messages by simulating a listener, does it learn robust pragmatics or just exploit that listener? We compare a mirror self-play selector, which chooses the candidate that succeeds with one fixed simulated listener, to a population-play selector, which chooses the candidate that succeeds across a small heterogeneous listener population. We then evaluate selected messages with held-out API listeners.

The paper contributes: (i) PRAG-CrossPlay, a released situated reference-game benchmark with procedural scenes, prompts, cached API traces, and verification scripts; (ii) a cross-play protocol that separates same-partner success from held-out communicative success; and (iii) a compact empirical diagnosis showing when self-play selection finds robust messages and when it instead selects partner-specific under-informative descriptions.

The current experiments support three findings. First, same-play success can overstate held-out communicative success: mirror self-play achieves perfect same-partner success but loses under cross-play. Second, the gap is concentrated in perspective-sensitive scenes, where short messages such as “the small blue sphere” or speaker-frame relational descriptions work for the training partner but fail for held-out listeners. Third, robust candidates are usually available, but selector design matters: population-play closes the observed gap when exact listener-invariant fallbacks are present, while a no-coordinate ablation shows that a consensus-plus-informativeness selector is needed when those fallbacks are removed.

2 Related Work

Interactive social-agent evaluation. SOTOPIA evaluates language agents in broad social role-play settings with goals such as coordination, exchange, and competition [Zhou et al. \(2023\)](#). SOTOPIA- π extends this line to interactive learning and reports that LLM-based evaluators can overestimate agents trained through social interaction [Wang et al. \(2024\)](#). PRAG-CrossPlay is narrower by design: it does not try to cover open-ended social intelligence, but gives an objective ground-truth success signal for one core communicative problem.

Cross-play and conventions. Self-play can produce conventions that work with a training partner but fail with independently trained or differently prompted partners. This concern is central in zero-shot coordination and motivates cross-play style evaluations such as Other-Play and Any-Play [Hu et al. \(2020\)](#); [Lucas & Allen \(2022\)](#), building on a broader view of conventions in communication [Lewis \(1969\)](#). Our contribution is to bring this diagnostic to natural-language reference games with API language-model listeners.

Reference games and pragmatics. Reference games are a standard testbed for pragmatic language use because the speaker must choose an utterance relative to the listener’s alternatives and context [Frank & Goodman \(2012\)](#); [Goodman & Frank \(2016\)](#). Recent scalable pragmatic communication work studies self-supervised utterance choice in reference-like settings [Hu et al. \(2021\)](#). PRAG-CrossPlay complements these lines by separating same-partner success from held-out listener success, making partner overfitting directly measurable.

3 PRAG-CrossPlay Task

Each PRAG-CrossPlay episode contains a grid scene, visible objects, a target object ID hidden from the listener, a speaker message, and a listener choice. Objects have color, shape, size, row, and column attributes. The listener sees the candidate objects but not the target label. The speaker is instructed not to mention the hidden ID and to use only listener-visible information.

We generate five scenario families. Unique-attribute scenes have targets that are identifiable by simple attributes. Distractor-contrast scenes include a distractor sharing salient

properties with the target. Relational-reference scenes require descriptions relative to a landmark. Perspective-shift scenes give the speaker and listener different orientations, making left/right and other frame-sensitive descriptions brittle. Partial-observability scenes give the speaker a private landmark that is hidden from the listener, testing whether messages use only listener-visible information. The main stress tests in this draft focus on perspective shift and partial observability because pilot results showed ceiling effects in the easier families.

Evaluation. For each held-out listener response, success is

$$\mathbb{1}[\text{choice.id} = \text{target.id}].$$

For self-play methods we also record same-play success under the listener used for selection. The cross-play gap is

$$\text{same-play success} - \text{held-out cross-play success}.$$

A positive gap indicates that same-partner success is overstating robustness.

4 Methods

API candidate generation. For each scene, an API speaker generates $K = 4$ candidate utterances. The prompt asks for a natural first attempt, an attribute-based message, a relational or spatial message, and an explicit fallback. The prompt includes the target attributes but not the target ID in candidate messages. All API calls are cached by prompt hash.

Baselines. The template baseline describes the target with attributes and exact row/column. The direct baseline uses the first API candidate. Best-of- K shortest selects the shortest generated candidate, controlling for candidate multiplicity without listener feedback.

Informativeness prior and consensus+info. In the no-exact-coordinate ablations, we also report a text-only selector that prefers candidates containing contrastive or spatial cues such as “only,” “leftmost,” “below,” or “next to.” We then combine this prior with local listener feedback: the consensus+info selector uses population feedback when a candidate has unanimous local support, and otherwise falls back to the informativeness prior. Neither selector uses held-out listener labels.

Mirror self-play. The mirror selector scores each candidate with a fixed simulated listener and chooses the shortest candidate that the listener maps to the target. This is a minimal model of same-partner communicative selection.

Population-play. The population selector scores each candidate across three heterogeneous local listeners. These listeners differ in tie-breaking, frame assumptions, and relation weighting. The selected message maximizes average success with a small length penalty.

Oracle candidate. For analysis only, we evaluate every generated candidate with held-out API listeners and report the best candidate’s held-out success. This measures whether failures come from candidate generation or candidate selection.

5 Experiments

We report three bounded API experiments. The mixed pilot contains 50 scenes from the four initial scenario families. The perspective-stress replication contains 50 new perspective-shift scenes. The partial-observability support run contains 50 private-landmark scenes. In all runs, API speakers generate candidates and held-out API listeners evaluate final messages. The selectors are local and deterministic, which keeps cost low and makes the partner-overfitting mechanism inspectable. We use paired bootstrap intervals over scenes for method differences.

Method	Cross-play	95% CI	Same-play	Gap
Template	1.000	[1.000, 1.000]	–	–
Direct	0.840	[0.747, 0.920]	–	–
Best-of-K shortest	0.740	[0.627, 0.853]	–	–
Mirror self-play	0.907	[0.833, 0.967]	1.000	0.093
Population-play	1.000	[1.000, 1.000]	1.000	0.000
Oracle candidate	1.000	[1.000, 1.000]	–	–

Table 1: Mixed 50-scene pilot. Mirror self-play has perfect same-play success but lower cross-play success; the gap is concentrated in perspective-shift scenes.

Scenario	Direct	Best-of-K	Mirror	Population	Oracle
Distractor Contrast	1.000	1.000	1.000	1.000	1.000
Perspective Shift	0.600	0.400	0.633	1.000	1.000
Relational Reference	0.733	0.533	0.933	1.000	1.000
Unique Attribute	1.000	1.000	1.000	1.000	1.000

Table 2: Mixed pilot by scenario type. The main degradation appears in perspective-shift scenes; unique-attribute and distractor-contrast scenes are near ceiling.

Reproducibility checks. All API responses are cached. The cache records exact response model versions: gpt-5.4-nano-2026-03-17 for the main API runs and gpt-4.1-nano-2025-04-14 for the alternate-model audit. We also include a generated protocol appendix with prompts and JSONL schemas, plus a benchmark integrity audit that checks generated messages for object-ID leakage, verifies listener choices and success flags against source scenes, and confirms balanced scene coverage across methods. A reviewer-facing artifact guide maps each main claim to the concrete source scenes, cached result files, tables, and verifier outputs used to support it. As a benchmark-scale artifact sanity check, the release also includes a no-API 600-scene balanced local sweep: mirror self-play has 1.000 same-play but 0.631 cross-play success, while population-play, template, and oracle candidate each reach 1.000.

6 Results

Table 1 shows that mirror self-play obtains 1.000 same-play success but 0.907 cross-play success in the mixed pilot. Population-play obtains 1.000 cross-play success on the same scenes. The paired population-minus-mirror difference is 0.093 with a 95% bootstrap interval of [0.027, 0.173].

Table 2 localizes the effect. Unique-attribute and distractor-contrast scenes are at ceiling. Relational-reference scenes show a smaller mirror gap. Perspective-shift scenes are the hard case: direct messages and shortest messages often under-specify the target, and mirror self-play selects messages that work for its fixed partner but fail under held-out listeners.

Table 3 confirms this pattern in 50 fresh perspective-shift scenes. Mirror self-play again achieves 1.000 same-play success but only 0.813 cross-play success. Population-play reaches 1.000 cross-play success, matching the oracle candidate upper bound. The paired population-minus-mirror difference is 0.187 with a 95% bootstrap interval of [0.100, 0.287].

Table 5 evaluates the same selected messages with an alternate held-out model, gpt-4.1-nano. The direct and shortest baselines become weaker, mirror self-play drops to 0.713 cross-play success, and population-play again reaches 1.000. The paired population-minus-mirror difference under this alternate held-out model is 0.287 with a 95% bootstrap interval of [0.193, 0.380].

Method	Cross-play	95% CI	Same-play	Gap
Template	1.000	[1.000, 1.000]	–	–
Direct	0.713	[0.600, 0.813]	–	–
Best-of-K shortest	0.420	[0.293, 0.540]	–	–
Mirror self-play	0.813	[0.720, 0.900]	1.000	0.187
Population-play	1.000	[1.000, 1.000]	1.000	0.000
Oracle candidate	1.000	[1.000, 1.000]	–	–

Table 3: Perspective-stress replication on 50 new scenes. Population-play matches the oracle candidate upper bound, while mirror self-play retains a substantial cross-play gap.

Scenario	N	Pop.	Mirror	Diff.
overall	50	1.000	0.813	0.187 [0.100, 0.287]
perspective shift	50	1.000	0.813	0.187 [0.100, 0.287]

Table 4: Paired bootstrap comparison of population-play versus mirror self-play on the perspective-stress replication.

Table 6 tests whether this result is merely a coordinate fallback effect. We remove one exact row/column candidate from each perspective-stress candidate set and recompute selection without new API calls. This ablation is deliberately stricter and exposes a limitation of the current local selector. Robust non-coordinate candidates remain available: the no-coordinate oracle reaches 0.793, and consensus+info reaches 0.760. However, naive population-play falls to 0.420 because the local training listeners often assign equal or low scores to the more informative non-coordinate descriptions, causing the length penalty to favor short ambiguous messages. Thus the headline population-play result should be read as robust fallback selection when explicit listener-invariant coordinates are available, while the ablation shows that selector quality matters when such fallbacks are removed. The same cache-only filter on the mixed pilot gives a no-coordinate oracle of 0.960 and consensus+info success of 0.920, with the gains over mirror concentrated in relational-reference and perspective-shift scenes.

Selection behavior explains the ablation. A candidate-role audit makes this concrete: in the full perspective-stress and partial-observability stress runs, population-play selects the coordinate-fallback slot in 1.000 of scenes; in the alternate-model perspective audit, this is 50 coordinate selections versus 17 for mirror self-play. Once exact coordinates are removed, naive population-play selects a spatially informative message in only 4 of 50 perspective-stress scenes and drops to 0.420 cross-play success. Consensus+info falls back to the informativeness prior in all 50 stress scenes, selecting spatially informative messages throughout and reaching 0.760; the same role audit shows it selects the short attribute-only slot in 0.000 of the no-coordinate perspective and partial-observability scenes. In the mixed no-coordinate pilot, consensus+info uses local consensus in 32 scenes and the informativeness fallback in 18 scenes.

The oracle result is important. In the perspective-stress replication, the oracle candidate succeeds on every scene, so the API speaker usually generated a robust message. The failure is selection: mirror self-play often selects a shorter message that is only partner-robust, whereas population-play selects the explicit fallback or a less frame-sensitive description.

A cache-only selection-regret audit makes this decomposition explicit. Regret is oracle candidate success minus method success, averaged by scene. In the full-candidate API runs, population-play has zero regret in the mixed, perspective-stress, alternate-model, and partial-observability settings, while mirror self-play leaves regrets of 0.093, 0.187, 0.287, and 0.333 respectively. In the no-coordinate ablations, consensus+info reduces regret to 0.040 in the mixed pilot, 0.033 in the perspective-stress audit, and 0.013 in partial observability, showing that most remaining failures are selector failures rather than missing candidates.

Held-out model	Direct	Best-of-K	Mirror	Population	Oracle
gpt-5.4-nano	0.713	0.420	0.813	1.000	1.000
gpt-4.1-nano	0.553	0.367	0.713	1.000	1.000

Table 5: Held-out listener model audit on the same 50 perspective-stress scenes and same API-generated candidates. The partner-overfitting gap persists when the held-out listeners use gpt-4.1-nano instead of gpt-5.4-nano.

Method	Cross-play	Same-play	Gap	Tokens
Direct first	0.553	-	-	11.7
Best-of-K shortest	0.367	-	-	4.6
Mirror self-play	0.580	1.000	0.420	8.3
Population-play	0.420	0.520	0.100	5.0
Consensus+info	0.760	-	-	12.5
Informativeness prior	0.760	-	-	12.5
Oracle candidate	0.793	-	-	9.4

Table 6: No-exact-coordinate ablation on the alternate-model audit. We remove candidate messages that explicitly mention row or column numbers, then recompute selection while reusing cached held-out listener evaluations.

202 An even stricter candidate-pool audit asks whether any candidate succeeds with every
 203 held-out listener in a scene. Every full-candidate run has at least one such robust candi-
 204 date in all 50 scenes, and population-play selects a robust candidate in 1.000 of scenes;
 205 mirror robust-selection rates are 0.860, 0.720, 0.560, and 0.500. After exact coordinates are
 206 removed, robust non-coordinate candidates remain available in 0.920, 0.660, and 1.000 of
 207 scenes for the mixed, alternate-model perspective, and partial-observability runs, and con-
 208 sensus+info matches the per-scene oracle in 0.940, 0.920, and 0.980 of scenes. A uniform-
 209 random candidate baseline rules out the explanation that any candidate would do: ran-
 210 dom expected success is 0.865, 0.715, 0.657, and 0.753 in the four full-candidate runs,
 211 so population-play improves over random by 0.135, 0.285, 0.343, and 0.247; in the no-
 212 coordinate runs, consensus+info improves over random by 0.100, 0.218, and 0.316. A
 213 message-length audit rules out a pure verbosity explanation. Population-play does se-
 214 lect longer messages than mirror self-play by 1.26, 2.82, 2.82, and 3.16 whitespace tokens
 215 on average in the four full-candidate runs, but direct messages are longer than mirror in
 216 the perspective and partial-observability runs while remaining less successful. Population-
 217 play messages also remain near the deterministic template scale: 6.82, 10.22, 10.22, and 9.66
 218 tokens on average, compared with a 10-token template baseline. A prefix-K candidate-
 219 budget audit shows why the four-candidate protocol is useful: in the alternate-model per-
 220 spective audit, robust-scene coverage rises from 0.380 at $K = 1$ to 0.660 at $K = 3$ and
 221 1.000 at $K = 4$; in partial observability, it rises from 0.620 at $K = 1$ to 1.000 at $K = 3$.
 222 A held-out listener-disagreement audit shows the same pattern at the selected message
 223 level. Mirror self-play produces split held-out outcomes in 0.080, 0.180, 0.360, and 0.500 of
 224 scenes across the full-candidate runs, while population-play has 0.000 split outcomes in all
 225 four. In no-coordinate partial observability, consensus+info reduces split outcomes from
 226 mirror’s 0.540 to 0.020. A listener-confidence audit further shows that self-reported un-
 227 certainty is not enough: in the alternate-model perspective runs, ambiguity flags are 0.000
 228 while mirror self-play still produces high-confidence failures (confidence ≥ 0.8) in 0.147 of
 229 full-candidate listener rows and 0.280 of no-coordinate listener rows.

230 To check that this is not only an artifact of the deterministic local selector, we also run
 231 a cache-only leave-one-listener-out analysis over the API all-candidate evaluations. For
 232 each scene, one API listener prompt is held out; API-mirror selection uses one different
 233 API listener prompt, while API-population selection uses the other two. Population se-
 234 lection improves over API-mirror selection in all four cached runs. The gains are small in
 235 the mixed and main perspective runs, but larger for the alternate-model perspective audit
 236 (0.940 vs. 0.847) and partial observability (0.940 vs. 0.767). Because this analysis reuses the

Setting	Gap	Same-play minus cross-play
Mixed 50, mirror	0.093	████
Mixed 50, population	0.000	
Perspective stress, mirror	0.187	████████
Perspective stress, population	0.000	

Figure 1: Cross-play gap for mirror and population selection. The gap is same-play success minus held-out cross-play success.

Failure set	Failures	Underspecified	Perspective	Other
Perspective alt-model mirror	43	38	5	0
Partial-observability mirror	50	50	0	0
Partial no-coordinate mirror	59	59	0	0
Combined	152	147	5	0

Table 7: Rubric-coded categories for mirror self-play held-out listener failures across the coded paper-facing failure sets. Each row count is a listener-level failure.

cached API listener pool for post-hoc splits, we treat it as a mechanism check rather than as the main held-out evaluation.

As an additional support check, we also run a 50-scene partial-observability stress check in which the speaker sees a private landmark hidden from the listener. The prompt instructs the speaker to use only listener-visible information. In this setting, mirror self-play obtains 0.667 cross-play success despite 1.000 same-play success, while population-play and the oracle both reach 1.000. After removing exact row/column candidates, robust non-coordinate messages remain available: the no-coordinate oracle is 1.000 and consensus+info reaches 0.987. The candidate audit finds zero explicit private-landmark references, so the observed failures again reflect under-informative messages that fit both the target and distractor rather than object-ID leakage or hidden-landmark leakage. Rubric coding supports this interpretation: all 50 full-run mirror failures and all 59 no-coordinate mirror failures are underspecified-distractor choices.

7 Qualitative Analysis

In scene ps_000005, the target is a small green sphere at row 4, column 1, with another small green sphere at row 1, column 4. Mirror self-play selects “It’s a small green sphere.” Its local partner resolves this correctly, but two of the three alternate-model held-out listeners choose the distractor. Population-play selects “The target is the small green sphere at row 4, column 1,” which all held-out listeners map to the target.

In scene ps_000011, mirror self-play selects “Find the large yellow cube left of the purple cylinder.” Under perspective shift, this relation is frame-sensitive: one alternate-model held-out listener chooses the same-shape same-color distractor. Population-play selects “Go to row 4, column 1: the large yellow cube,” which all held-out listeners map to the target.

Table 7 consolidates the rubric-coded mirror failures used in the analysis. Across 152 listener-level failures, 147 are underspecified distractor cases and 5 are perspective-frame errors; no coded failures are wrong-attribute, private-landmark, or listener-misparse cases. A generated qualitative appendix adds two partial-observability examples, including a no-coordinate consensus+info repair that raises held-out success from 0.333 to 1.000 in the selected scene.

267 8 Limitations

268 The current experiments are intentionally small and diagnostic. The selector partners are
269 local deterministic listeners rather than trained LLM agents. This is useful for isolating
270 partner overfitting, but it means the result should be framed as a cross-play diagnostic
271 rather than as evidence about large-scale agent training. The held-out listeners are API
272 language models, not humans. We audit two held-out model families, but both are still
273 API LLMs and prompt variants rather than independent human partners. The strongest
274 population-play result relies on exact coordinate fallbacks; without those fallbacks, the cur-
275 rent local listener population is not sufficient, although consensus+info recovers much of
276 the no-coordinate oracle. The grid scenes are simplified and text-rendered, so the bench-
277 mark tests situated reference rather than embodied perception. Finally, the strongest evi-
278 dence is concentrated in two diagnostic families, perspective shift and partial observability;
279 broader claims require larger balanced runs across all scenario types.

280 9 Conclusion

281 Same-partner communicative success is not sufficient evidence of robust pragmatic com-
282 munication. In PRAG-CrossPlay, mirror self-play can achieve perfect same-play success
283 while failing under held-out listeners, especially when messages are perspective-sensitive
284 or under-specified. Population-play mitigates this failure when listener-invariant fallbacks
285 are available, but the no-coordinate ablation shows that cross-play is also useful for diag-
286 nosing and repairing weaknesses in the selector itself. These results support cross-play as a
287 simple, objective evaluation axis for language agents that learn or select messages through
288 situated interaction.

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